Group Recommender Systems for board games: performance and aggregation strategies

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Abstract

Group recommendation is a challenging task for recommender systems, as it requires satisfying multiple users simultaneously. This work evaluates the following models: Random, Most Popular, Matrix Factorization (MF), MF for Groups, SVD++, and DeepFM on the BoardGameGeek dataset for board games with four or more players. Three aggregation strategies (Average, Least Misery, Most Pleasure) were applied to combine individual preferences into group recommendations.

Results show that Matrix Factorization achieves the best balance between relevance (Recall@10) and ranking accuracy (nDCG@10), while DeepFM stands out in novelty, enabling the discovery of niche games. MF for Groups shows a slight decrease in recall due to smoothing extreme preferences but remains competitive and offers higher computational efficiency by producing a single prediction per group. Aggregation strategy has a marginal effect compared to the choice of base model.

These findings show how important the base algorithm is, how hybrid models help explore new or niche games, and how latent factor models consistently provide relevant and satisfying recommendations for groups.

1. Introduction

Recommender systems are a fundamental component of digital platforms, helping users navigate vast catalogs through personalized suggestions based on the content they interact with. However, most systems focus on individual recommendations, even though—as social beings—many of our daily decisions are made in group contexts: choosing a restaurant to dine with friends, selecting a movie to watch with a partner, or deciding which board game to play on a Saturday afternoon. These situations require merging different preferences into a single recommendation, introducing the challenge of constructing collective preferences that resolve conflicts and ensure fairness among group members.

The task of selecting a board game is particularly valuable for group recommendations, as most available options involve interactions among multiple people and their individual preferences. The enjoyment a game provides depends on the compatibility between its characteristics and the personal tastes of each participating player. BoardGameGeek offers a large database with user ratings and extensive information about each game (mechanics, categories, designers, textual descriptions). This multimodality enables exploring recommendations based both on user ratings and game content. In this paper, we use this dataset to analyze group recommendations for board games.

To generate a group recommendation, it is necessary to apply aggregation strategies, which combine individual preferences, resolve potential conflicts among them, and construct a recommendation that is acceptable to all group members. Several types of strategies exist, but for this work we focus on the following: Average, which computes the mean of the expected ratings; Least Misery, which focuses on the least enthusiastic user; and Most Pleasure, which maximizes the preference of the most enthusiastic user. These strategies were selected to cover the three main situations one faces when making a group decision. Average reflects a group-level consensus, Least Misery ensures that no member hates the chosen option—thus encouraging everyone's participation—while Most Pleasure prioritizes the most enthusiastic user to promote engagement with the game. This allows us to analyze how each prioritization impacts overall satisfaction and the quality of the resulting group ranking.

Although more advanced approaches such as CubRec or ConsRec model recommendations as distributed regions rather than single points, they require higher computational complexity and additional data that this dataset does not provide. Therefore, they are not considered in the present work.

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To generate the recommendations, we implemented two baseline models (Random and Most Popular) and a hybrid model (DeepFM). For the Matrix Factorization (MF) models, this study contrasts two different architectural approaches: (1) Aggregated Predictions (MF), where individual scores are combined after prediction, and (2) Aggregated Models (MF for Groups), where latent group profiles are constructed before training, allowing us to analyze the trade-off between accuracy and computational efficiency.

The findings show that matrix factorization and hybrid models achieve strong performance at the individual level. At the group level, performance remains consistent across aggregation strategies, suggesting that the choice of base model has a greater impact than the consensus rule itself. Under absolute agreement (all users rate the item positively), Most Popular outperforms the other models.

Our contributions are: (1) a comprehensive evaluation of baseline, matrix factorization, and hybrid models on a diverse, large-scale dataset; (2) an empirical comparison of three aggregation strategies (Average, Least Misery, Most Pleasure) and two group-modeling architectures (Aggregated Predictions vs Aggregated Models); and (3) a technical analysis of group-modeling architectures, contrasting the effectiveness of aggregating individual predictions with the construction of latent group profiles (Aggregated Models). This paper explores the opportunities and challenges of group recommendations on a multimodal dataset, offering practical insights and methodological considerations for future research.

2. State of Art

Recommender systems aim to estimate personalized preferences. In group contexts—where recommendations must satisfy multiple users simultaneously—challenges arise such as aggregating heterogeneous preferences, ensuring fairness among members, and maximizing collective satisfaction (Jameson & Smyth, 2007). Group Recommender Systems (GRS) address these aspects through two main approaches (Boratto & Carta, 2014; Felfernig et al., 2018):

→ **Aggregation of individual predictions:** Independent recommendations are generated for each user using individual models (MF or hybrid models), and the predictions are then combined using strategies such as: Average (averages ratings to seek overall balance), Least Misery (takes the minimum rating, prioritizing the avoidance of items strongly disliked), and Most Pleasure (selects the maximum rating, maximizing the satisfaction of the most enthusiastic member) (Masthoff, 2015). This method is computationally efficient but may overlook more complex group dynamics.

→ **Group profiles:** This approach integrates individual preferences into a collective virtual user prior to the training phase. This allows the model to learn latent patterns directly at the group level, although it risks smoothing out extreme preferences of individual members (Baltrunas et al., 2010).

Matrix Factorization (MF) is a foundational approach that represents users and items in a shared latent space, capturing implicit patterns to enable accurate predictions (Felfernig et al., 2014). Models such as SVD and SVD++ extend this idea by incorporating biases and additional factors, improving performance on sparse datasets (Koren et al., 2009). Hybrid models like DeepFM integrate collaborative filtering with content-based features, strengthening robustness in multimodal domains (Guo et al., 2017). Recent advances include approaches based on distributed regions (CubRec, ConsRec), which model preferences as probabilistic distributions (Wu et al., 2023), although they are computationally demanding and require contextual data that is not always available. Recent studies also highlight the incorporation of bias terms to refine aggregation methods (Koren et al., 2009; Jameson & Smyth, 2007) and the expansion of GRS towards group decision support (Delić et al., 2024).

This work adopts individual prediction aggregation as the primary approach, applying this strategy to the baseline models, DeepFM, and the standard MF version. However, in order to contrast different architectures, we additionally implement the group-profile strategy (Aggregated Models) specifically within the Matrix Factorization setting. This allows us to isolate the impact of the aggregation methodology (pre-training vs post-prediction) while keeping the base algorithm (SVD) constant, evaluating its performance in terms of relevance, novelty, and diversity (Nguyen et al., 2024).

3. Dataset

The dataset used was the Board Games Database from BoardGameGeek, available on Kaggle (Wadkins, 2021). It originally contains information for approximately 22,000 board games with ratings from 1 to 10, more than 411,000 users, and a total of 19 million reviews. This dataset is multimodal, including explicit ratings and detailed item information (mechanics, categories, designer), which enables the exploration of hybrid approaches.

To ensure training quality and reduce sparsity, we applied an iterative filtering preprocessing step (k-core filtering), retaining only users and items with at least 5 interactions. Descriptive analysis of the final dataset reveals a rating distribution with a positive skew (J-shaped), as shown in Figure 1, as well as a concentration in a popular subset of items, with many games receiving only a small number of reviews, as illustrated in Figure 2.

Due to the absence of explicit information about group interactions in the platform, we adopted a protocol for generating

synthetic groups. Groups of fixed size (N=4) were formed through random sampling of users from the validation set, simulating casual decision-making scenarios among heterogeneous users.

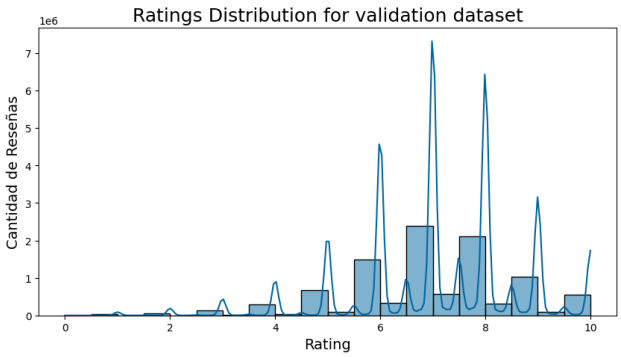

Figure 1. Number of reviews vs rating for the validation dataset.

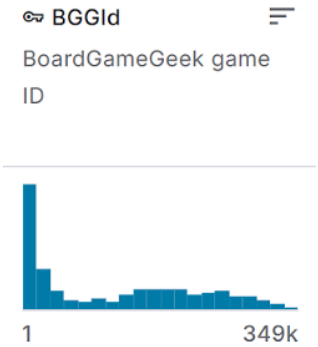

Figure 2. Distribution of board game IDs.

4. Methodology

The methodology addresses the group recommendation problem through a comparative experimental design. Two fundamental aggregation architectures are contrasted:

→ **Aggregated Predictions:** Individual recommendations are generated and then merged using consensus strategies (applied to Baselines, MF, and DeepFM).

→ **Aggregated Models:** Latent group profiles are constructed before training (applied exclusively to MF to evaluate efficiency and potential loss of information).

Six models were evaluated, covering different paradigms: Random (minimum expected performance), Most Popular (real platform behavior and dataset biases), Matrix Factorization and MF for Groups (discovers latent relationships while mitigating sparsity), SVD++ (extends MF with implicit feedback), and DeepFM (integrates collaborative filtering with metadata, catching higher-order interactions).

4.1. Dataset Preprocessing

Interactions were binarized by considering ratings $r \geq 7$ as positive, aligned with the dataset's average rating (7.1) and the BGG standard of a "Good game." This facilitates ranking metrics and reduces noise. An iterative filter (k-core) was applied, retaining users and items with at least 5 interactions to mitigate sparsity. These characteristics highlight the inherent sparsity of the dataset, justifying the use of models such as MF and DeepFM to capture latent patterns. MF and SVD++ were implemented with Surprise, while DeepFM was trained using DeepCTR-Torch, incorporating mechanics and categories as feature inputs. Due to its computational cost, DeepFM was trained on 1 million stratified interactions, evaluating its generalization under moderate resource constraints.

For preprocessing, 60% of the data was allocated for training and the remaining 40% split into validation and testing. Before performing this split, games were filtered according to their maximum allowed number of players. Consequently, a minimum threshold of 4 players was established for a game to be considered valid. This decision offers a balance between the amount of available information and the level of social interaction. As shown in Table 4.1, when considering games that allow at least 4 players, a large proportion of total ratings is preserved (16.6 million out of 18.9 million), while still focusing the analysis on a group-based context. If the minimum number of players is set to 5 or 6, the number of available games and interactions drops significantly.

No. of players	No. of games	No. of ratings
2+ players	21387	18807316
3+ players	17035	16686750
4+ players	16748	16582133
5+ players	9880	9519795
6+ players	6836	5405768

Table 1. Number of games and ratings according to the number of players.

The table 4.1 presents the statistics of the complete dataset and compares them with the training and validation datasets obtained after filtering.

Individual predictions were merged using three classical strategies: Average (averages preferences, maximizing overall satisfaction), Least Misery (selects the minimum rating, preventing strong dissatisfaction), and Most Pleasure (selects the maximum rating, prioritizing specialized or enthusiastic preferences). These strategies represent different real-world mechanisms: Average balances individual differences; Least Misery is relevant in settings where a bad experience affects the entire group; Most Pleasure reflects

Statistic	Full	Train	Val
# Users	398517	398517	250410
# Items	16748	16748	16744
# Ratings	13414601	10200445	3202256
Avg. Ratings/User	33.66	25.60	12.79
Avg. Ratings/Item	800.97	609.05	191.25
Avg. Rating	7.14	7.15	7.10
Std. Rating	1.54	1.54	1.53
Max. Ratings/User	4413	3310	1103
Max. Ratings/Item	88254	68263	20001
Min. Ratings/User	1	1	1
Min. Ratings/Item	6	6	1
Density (%)	0.20	0.15	0.08

Table 2. Descriptive statistics of the data.

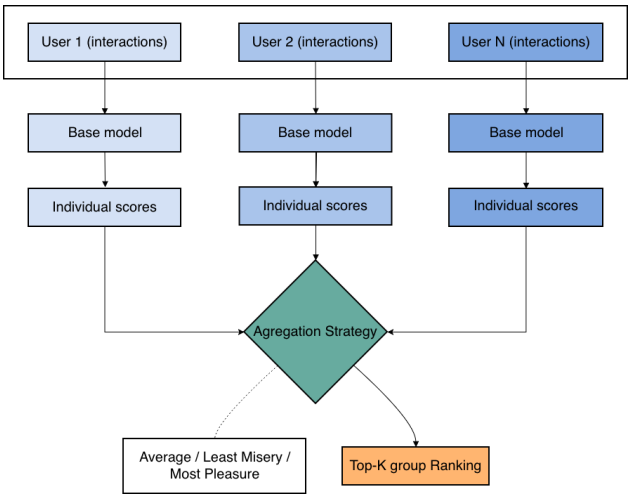

Figure 3. Aggregation of individual predictions.

dynamics where an expert or highly enthusiastic member guides the decision.

For the evaluation of group recommendations, 1,000 synthetic groups of size N=4 were generated by randomly sampling users from the validation set, under the constraint that all group members shared at least one commonly rated item. This criterion ensures a minimal shared information basis for preference aggregation, without assuming previous compatibility among the members, thus allowing us to evaluate aggregation strategies under realistic conditions where preferences differ. The choice of N=4 takes into consideration the previously applied dataset filter and reflects a representative case of social interaction in board game contexts. Figure 3 presents a conceptual overview of the methodology used.

The evaluations used Precision@10, Recall@10, and nDCG@10 in both individual and group settings. For the group setting, a strict consensus criterion was adopted: an item is considered relevant only when all members rate it with r $\geq$ 7, a threshold aligned with the dataset's global mean rating (7.1), replicating scenarios of non-delegated collective satisfaction.

4.2. Parameter analysis

The hyperparameters were selected through empirical experimentation, considering standard values from the literature and adjustments based on validation performance. Sensitivity was evaluated for latent factors, training epochs, and learning rates to balance precision, recall, and computational complexity. The binarization threshold (r $\geq$ 7) was aligned with the dataset mean (7.1) and the BGG standard.

4.2.1. Baselines (Random and Most Popular)

These models do not require complex hyperparameter tuning. Random serves as a lower-bound baseline, while Most Popular is based on global popularity with no tuning. In general, hyperparameters were optimized to maximize the trade-off between relevance (precision/recall) and exploration (novelty/diversity), with particular emphasis on nDCG@10 for ranked recommendations.

4.2.2. Matrix Factorization–Based Models (MF, MF for groups and SVD++)

For MF, MF for groups, and SVD++ (implemented using Surprise), the following hyperparameters were used: $n_{factors} = 100$ (to capture complex patterns in sparse data), $n_{epochs} = 20$, $lr_{all} = 0.005$ for MF and MF for groups, and $lr_{all} = 0.007$ for SVD++, and $reg_{all} = 0.02$ (to prevent overfitting). Hyperparameter tuning was performed via cross-validation on approximately 10% of the data, prioritizing nDCG@10. SVD++ incorporates implicit feedback, making it more robust to sparsity at the cost of higher computational complexity.

4.2.3. DeepFM

For DeepFM (implemented with deepctr-torch), the following settings were used: dnn_hidden_units=(128,128), dnn_dropout=0.5 (to mitigate overfitting on multimodal features), l2_reg_dnn=0.001 (to balance generalization), 10 epochs with batch_size=2048, and the task set to 'binary' to match the rating binarization. Due to computational constraints, training was performed on 1M sampled interactions, prioritizing diversity.

5. Experiments and Results

The experiments were conducted on the BoardGameGeek dataset, applying the five models under the protocol described in the Methodology section. We evaluated Preci-

sion@10, Recall@10, nDCG@10, Novelty, and Diversity to capture both relevance and exploratory capacity.

5.1. Individual Results

In the individual scenario shown in Table 3, Most Popular dominates in precision metrics (0.4750) and nDCG (0.8374), an expected outcome given the strong popularity bias in the BoardGameGeek dataset. However, Matrix Factorization (MF) drastically outperforms all other models in Recall@10 (0.8673 vs 0.8404 for Most Popular), demonstrating its ability to retrieve relevant items in the long tail, which popularity-based models tend to overlook. DeepFM stands out notably in Novelty (18.70), suggesting that the inclusion of metadata (mechanics, categories) enables the model to recommend "niche games" that other models ignore—even though this comes at the cost of the Precision metric.

Model	P@10	R@10	nDCG@10	Novelty	Diversity
Random	0.0893	0.7457	0.1661	11.3240	0.1879
MP	**0.4750**	0.8404	**0.8374**	10.6201	0.7372
MF	0.2057	**0.8673**	0.5251	10.9902	0.4944
DeepFM	0.4609	0.8299	0.8214	**18.7068**	0.5446
SVD++	0.4680	0.8358	0.8298	10.7155	**0.7400**

Table 3. Individual recommendation performance. MP = Most Popular, MF = Matrix Factorization (SVD++), P@10 = Presicion@10, R@10 = Recall@10.

5.2. Group Results

In the group scenario shown in Table 4, two dimensions are analyzed: consensus strategies and modeling architectures.

→ **Aggregation Strategies:** When comparing Average, Least Misery and Most Pleasure, differences in ranking metrics are marginal (variations $\leq 1\%$). This suggests that, for this dataset, the choice of base model has a much greater impact on the final quality than the mathematical rule used to combine the scores.

→ **Comparison of Architectures (MF vs MF for Groups):** A key phenomenon is observed when contrasting Matrix Factorization (Aggregation of Predictions) against MF for Groups (Model/Profiles Aggregation).

- The classical approach (aggregated individual predictions) consistently achieves higher Recall (0.25 vs 0.21).
- The MF for Groups approach exhibits a slight degradation in performance. This supports the "information loss" hypothesis: by averaging user profiles before training, extreme preferences are smoothed out, making it harder for the latent model to capture specific individual tastes.
- Nevertheless, MF for Groups remains competitive ($nDCG \geq 0.96$) and represents a viable alternative when computational efficiency at inference time is required since it produces a single prediction per group instead of N individual predictions.

AVERAGE AGGREGATION					
Model	**P@10**	**R@10**	**nDCG@10**	**Novelty**	**Diversity**
Random	0.4200	0.1883	0.6284	13.6838	**0.9284**
Most Popular	0.8999	0.1452	0.9675	10.0912	0.8058
MF	0.8958	**0.2578**	**0.9726**	10.6313	0.8394
DeepFM	0.8537	0.1365	0.9456	**19.0039**	0.7699
SVD++	**0.9022**	0.1434	0.9656	10.4663	0.8399
MF for Groups	0.8848	0.2155	0.9610	10.3867	0.8303
LEAST MISERY AGGREGATION					
Random	0.4200	0.1883	0.6284	13.6838	**0.9284**
Most Popular	0.8999	0.1452	0.9675	10.0912	0.8058
MF	0.8965	**0.2580**	**0.9727**	10.6429	0.8397
DeepFM	0.8551	0.1367	0.9460	**19.0074**	0.7707
SVD++	**0.9007**	0.1431	0.9654	10.4878	0.8402
MF for Groups	0.8850	0.2156	0.9585	10.4766	0.8318
MOST PLEASURE AGGREGATION					
Random	0.4200	0.1883	0.6284	13.6838	**0.9284**
Most Popular	0.8999	0.1452	0.9675	10.0912	0.8058
MF	0.8907	**0.2563**	**0.9707**	10.6193	0.8396
DeepFM	0.8518	0.1362	0.9432	**18.9998**	0.7764
SVD++	**0.9090**	0.1445	0.9677	10.3857	0.8391
MF for Groups	0.8832	0.2152	0.9589	10.3136	0.8303

Table 4. Group recommendation performance under three aggregation strategies. P@10 = Presicion@10, R@10 = Recall@10.

6. Results Analysis

The obtained results allow us to draw relevant conclusions about the behavior of the different models and the influence of aggregation strategies on the group recommendation problem.

6.1. Individual Performance

At the individual level, **Matrix Factorization (MF)** demonstrates superior retrieval capability, achieving the highest **Recall@10** (0.8673), slightly outperforming **SVD++** (0.8358) and **DeepFM** (0.8299). This high performance suggests that latent factor models are effective at retrieving relevant items in the "long tail", beyond the extremely popular titles.

On the other hand, the **Most Popular** baseline marginally leads in **Precision@10** (0.4750) and **nDCG@10** (0.8374). This indicates that, in this dataset, popularity acts as a strong predictor of general acceptance, although it carries the risk of reducing diversity and generating redundant recommendations. In contrast, **DeepFM** stands out for its exploratory capability, achieving the highest **Novelty** (18.70), indicating that the integration of metadata (mechanics, categories) allows the model to recommend niche games that other models overlook, highlighting a clear trade-off between immediate precision and discovery.

6.2. Group Results and Aggregation Strategies

In the group scenario, performance differences between models remain consistent with the individual case. However, a methodologically relevant finding emerges: the three aggregation strategies evaluated (**Average, Least Misery, Most Pleasure**) show extremely similar metrics. For instance, under the MF model, **nDCG@10** is 0.9726 for **Average** and 0.9727 for **Least Misery**, showing practically no variation.

This demonstrates that the algorithm (MF, DeepFM, etc.) is the main determinant of performance, while the mathematical aggregation method acts as a marginal perturbation. This result partially contrasts with previous literature on Group Recommender Systems (GRS), where more pronounced effects of the consensus rule are typically observed in domains that are more sensitive to preference polarization.

Regarding group exploration, DeepFM maintains its advantage, achieving the highest Novelty (19.44 under the **Most Pleasure** strategy), substantially outperforming MF (10.91) and Most Popular (10.47). This positions hybrid models as particularly beneficial for groups that value exploration of specific themes, although this comes with a decrease in Recall@10 compared to MF.

6.3. Comparison of Architectures: Predictions vs Aggregated Models

A key phenomenon is observed when contrasting the two implemented Matrix Factorization architectures: MF with Aggregated Predictions versus MF for Groups (Aggregated Models).

The classical approach (aggregated individual predictions) consistently achieves higher Recall@10 (0.25) compared to the aggregated models approach (0.21). This degradation in the performance of MF for Groups supports the "information loss" hypothesis: by averaging user profiles and creating a group user before training, extreme tastes and specific preferences are smoothed, making it more difficult for the latent model to capture individual nuances.

However, **MF for Groups** remains competitive in terms of ranking (nDCG ≥ 0.96) and represents a viable alternative in scenarios where computational efficiency is critical, as it requires producing a single prediction per group instead of N individual predictions.

7. Conclusions

This study addressed the challenge of group recommendation in the domain of board games, an inherently social context where collective satisfaction depends on reconciling heterogeneous preferences. Through an extensive evaluation on the BoardGameGeek dataset, we compared multiple modeling paradigms and aggregation strategies, drawing three key conclusions that inform the design of future Group Recommender Systems (GRS).

First, we demonstrated that the choice of the base algorithm has a significantly greater impact on recommendation quality than the aggregation strategy employed. Performance differences between Average, Least Misery, and Most Pleasure were marginal compared to the differences between models (MF, DeepFM, Baselines). Specifically, Matrix Factorization (MF) emerged as the most balanced solution, offering the best trade-off between retrieving relevant items (Recall) and achieving accurate ranking (nDCG), outperforming approaches based purely on popularity or content.

Second, our architectural comparison between Aggregated Predictions and Aggregated Models (MF for Groups) revealed a key technical finding: a previous aggregation of user profiles leads to information loss that slightly degrades Recall (dropping from 0.25 to 0.21) due to the smoothing of extreme preferences. However, we confirmed that the Aggregated Models architecture remains competitive in high-demand scenarios, offering superior computational efficiency by producing a single prediction per group instead of multiple individual predictions.

Finally, we identified clear roles for each model depending on the usage context. While hybrid models like DeepFM proved superior for promoting Novelty and discovering niche games, Matrix Factorization under the Least Misery strategy emerges as the safest option to prevent conflicts in groups with divergent tastes.

As future work, we propose exploring aggregation mechanisms that dynamically weight each member's influence based on their expertise or satisfaction history, as well as incorporating textual explanations to justify to the group why a specific game satisfies their collective constraints.

Accessibility

All codes used in this paper are available in: https://github.com/Vicco02/Proyecto-SysRec-11

References

Baltrunas, L., Makcinskas, T., and Ricci, F. Group recommendations with rank aggregation and collaborative filtering. In *Proceedings of the Fourth ACM Conference on Recommender Systems (RecSys '10)*, pp. 119–126. ACM, 2010. doi: 10.1145/1864708.1864733.

Boratto, L. and Carta, S. State-of-the-art in group recommendation and new approaches for automatic identification of groups. In *Advances in Intelligent Information Hiding and Multimedia Signal Processing*, pp. 1–20.

Springer, 2014. doi: 10.1007/978-3-319-50212-0_1.

Delić, A., Masthoff, J., and Neidhardt, J. Group recommender systems: Beyond preference aggregation. In Ricci, F., Rokach, L., and Shapira, B. (eds.), *Recommender Systems Handbook*, pp. 1–40. Springer, 3rd edition, 2024. doi: 10.1007/978-1-0716-2197-4_19-1.

Felfernig, A., Boratto, L., Stettinger, M., and Tkalčič, M. Matrix factorization for group recommendation. In *Group Recommender Systems: An Introduction*, chapter 2. 2014. URL https://ase.sai.tugraz.at/wp-content/uploads/sites/34/2014/01/grouprecommendersystemschapter2.pdf.

Felfernig, A., Boratto, L., Stettinger, M., and Tkalčič, M. *Group Recommender Systems: An Introduction*. Springer, 2018. doi: 10.1007/978-3-319-75067-5.

Guo, H., Tang, R., Ye, Y., Li, Z., and He, X. Deepfm: A factorization-machine based neural network for ctr prediction. *arXiv preprint*, arXiv:1703.04247, 2017. doi: 10.48550/arXiv.1703.04247.

Jameson, A. and Smyth, B. Recommendation to groups. In Brusilovsky, P., Kobsa, A., and Nejdl, W. (eds.), *The Adaptive Web: Methods and Strategies of Web Personalization*, pp. 596–627. Springer, 2007. doi: 10.1007/978-3-540-72079-9_20.

Koren, Y., Bell, R., and Volinsky, C. Matrix factorization techniques for recommender systems. *IEEE Computer*, 42(8):30–37, 2009. doi: 10.1109/MC.2009.263.

Masthoff, J. Group recommender systems: Aggregation, satisfaction and group attributes. In Ricci, F., Rokach, L., and Shapira, B. (eds.), *Recommender Systems Handbook*, pp. 743–776. Springer, 2015. doi: 10.1007/978-1-4899-7637-6_22.

Nguyen, L. V., Vo, T. H., Thai, T. H., Nguyen, G. H., and Lee, O. J. A survey of recommendation systems: Datasets, evaluation methods, and application domains. *Applied Artificial Intelligence*, 38(1), 2024. doi: 10.1080/08839514.2024.2373873.

Wadkins, J. Board games database from boardgamegeek, 2021. URL https://www.kaggle.com/datasets/threnjen/board-games-database-from-boardgamegeek. Kaggle Dataset.

Wu, X., Zhang, F., Zhao, Z., Li, Y., Zhang, J., Wu, Y., and Guo, Y. Consrec: Learning consensus behind interactions for group recommendation. In *Proceedings of the ACM Web Conference 2023 (WWW '23)*, pp. 240–250. ACM, 2023. doi: 10.1145/3543507.3580887.